Higher Education Costs and Affordability in China and the United States

An exploratory comparative study using public data

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October 2026

Abstract

How expensive is college, and how can we compare college costs in different countries? I studied in China from primary school through senior high school. I am studying at Santa Monica College in the United States now. Therefore, I have come to believe that tuition fees are not the only expenses for families to study at university. I have obtained the following public data for 2025-2026: U.S. college costs, tuition policies and university fees in Shanghai, income indicators, and purchasing-power-adjusted GDP per capita. According to the data, published US tuition fees differ significantly for different schools and residents. Estimated housing and food expenses at public two-year colleges are about 72% of the total tuition, housing and food costs considered here. Tuition-to-income ratios provide the background, but the data in China and the US are not the same. Due to the above reasons, I cannot determine which country has more affordable colleges. Otherwise, I will present what we can learn from the data and what information is still lacking in other studies.

Higher Education Costs and Affordability in China and the United States

I went to primary and secondary school in China. I study at Santa Monica College in California now. After studying in both countries, I became more curious about the reasons for the different college costs. Tuition and living costs are frequently mentioned by people in the US. The college cost information I saw in China was usually shown differently. I wish to learn the content of these figures.

A high-tuition university is not necessarily out of reach for students from poorer families. Income level, living environment and financial support vary in different places and schools. Even in the same country, students in different states may have different tuition fees. There are two problems in this study: How different are the published costs of U.S. colleges? What information is needed to compare the affordability of colleges in China and the United States fairly?

Publicly available data will be used to compare college expenses in this paper. Only the Shanghai part is shown in the China section, not the whole country. I do not need to list the reasons for the difference in tuition or support a particular educational model.

Background and Related Research

College affordability is not the same. The list price of tuition fees at the university before grants and scholarships. Net price is the amount a student has to pay after grant subsidies. The total expenses for the student budget also cover housing, food, books, transportation, etc. These measures cannot be substituted. The College Board publishes U.S. college prices and student aid data, and the Organisation for Economic Co-operation and Development (OECD) investigates how countries finance higher education (College Board, 2025; OECD, 2026).

Income is also a reason; however, there is no single indicator of it. Per capita income is the average income of the people living in a place. A low-income family or a family with many children will have a very small budget. Purchasing Power Parity (PPP) accounts for the difference in price levels among countries, but PPP-adjusted GDP per capita does not show how much money a family has available for tuition (World Bank, 2024).

Therefore, although tuition fees are often used as a general measure, they cannot be employed to judge the economic level of different countries. First, learn what all these numbers are and to whom they belong.

Data and Methods

Data Sources

I obtained the public report and data from the Internet. Table 1 shows the source of the data, contents of the data, and omissions in the data.

Table 1

Public Data Sources and Their Limits

Source	Year and area	Measure and limitation
College Board, Table CP-1	2025–26, United States	Average published tuition and estimated housing/food by college sector; not individual net price
Shanghai tuition policy	From 2023, Shanghai	Annual public-university base tuition ceilings by program; not an average paid
Shanghai Statistics Bureau	2025, Shanghai	Per-capita resident disposable income; city-level income context
BEA via FRED	2025, United States	Per-capita disposable personal income; national measure with a different definition
World Bank WDI via UNdata	2024, China and US	PPP GDP per capita; economic output, not household cash income

Note. Sources are the cited reports and official datasets. The measures have different coverage and definitions.

Calculations

First, I added published tuition and fees to the estimated housing and food expenses for each group of U.S. colleges. Then, I divided the total by this amount and multiplied by 100 to obtain the proportion of tuition fees.

Second, I calculated an exploratory tuition-to-income ratio by dividing the published tuition fee by the annual per capita income of the area in question. Report the results from Shanghai and the U.S. separately due to different data definitions. I do not believe that the percentages can tell us which country's families are doing better financially for college. Currency conversion cannot address the difference in living expenses and school fees of different regions.

What This Method Cannot Measure

These data are not individual student records. They cannot show the amount of grant money a student has received, whether the student lives with family, or how the household income is distributed among expenses. To present patterns and raise new questions, do not predict whether a certain family can afford university education.

Results

U.S. College Type Changes the Cost Picture

According to the College Board (2025), the average published tuition and fees for 2025-26 were $4,150 at public two-year colleges and $45,000 at private non-profit four-year colleges. After adding the estimated housing and food, the total amounts were $15,000 and $60,920. The average in-state

tuition for public four-year universities is $11,950; out-of-state students pay about $31,880. There is a difference in tuition fees for students with different places of study at the same college.

Table 2

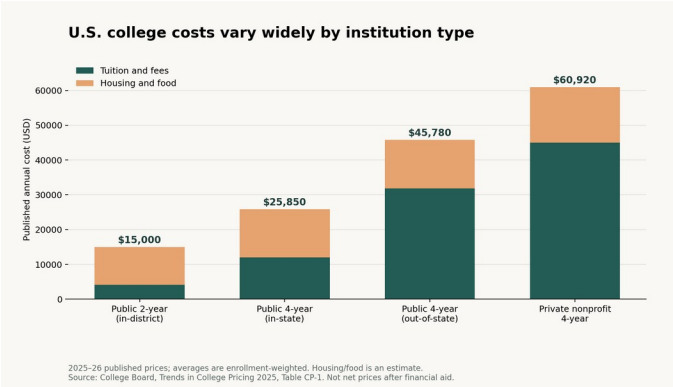
Published U.S. Tuition, Housing and Food by College Type, 2025–26

U.S. institution type	Tuition and fees	Housing and food	Combined amount	Tuition share
Public 2-year, in-district	$4,150	$10,850	$15,000	27.7%
Public 4-year, in-state	$11,950	$13,900	$25,850	46.2%
Public 4-year, out-of-state	$31,880	$13,900	$45,780	69.6%
Private nonprofit 4-year	$45,000	$15,920	$60,920	73.9%

Note. Amounts are in U.S. dollars. Tuition shares are calculated from College Board (2025), Table CP-1; totals exclude some other student expenses.

Figure 1

Published U.S. Tuition and Estimated Housing and Food, 2025–26



Note. Data from College Board (2025), Table CP-1. Amounts are sector averages, not individual bills.

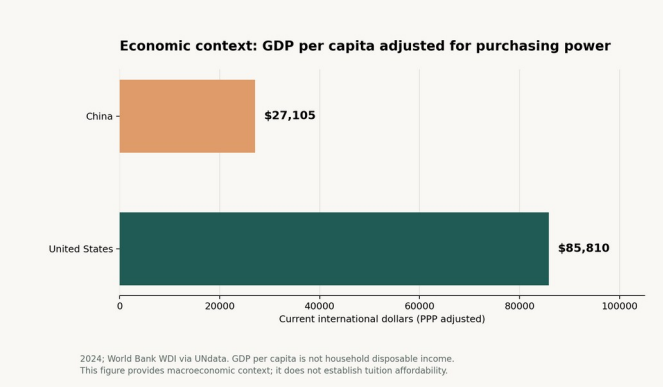
As a community college student, I have been learning about the numbers of two-year colleges. Tuition accounts for only 27.7% of the total of tuition, housing and food; housing and food together account for about 72.3%. It does not mean all community college students have the same cost of living. Some students live in rented rooms; others live with their families. However, according to the above statistics, tuition fees are not the only expenses a student will have.

Income and the Economic Setting

According to the Shanghai Statistics Bureau (2026), Shanghai's annual per-capita resident disposable income was 91,987 yuan in 2025. In the United States, per-capita disposable personal income was $67,221 in 2025 (U.S. Bureau of Economic Analysis, 2026. Both are relative to some statistics and different in terms of geography. In 2024, PPP-adjusted GDP per capita was about 27,105 international dollars in China and 85,810 in the United States, a ratio of about 3.17 to 1 (World Bank, 2024). It is not the income a family can use to pay for college, but economic output per capita.

Figure 2

GDP per Capita Adjusted for Purchasing Power, 2024



Note. Data from World Bank World Development Indicators, accessed through UNdata. Values are in current international dollars.

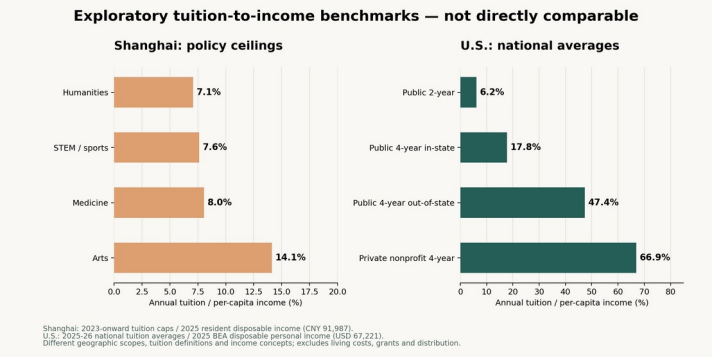
Exploratory Tuition and Income Ratios

Shanghai's base tuition ceiling for science and engineering programs is 7,000 yuan per year, about 7.6% of Shanghai's 2025 per-capita resident disposable income (Shanghai Municipal Development and Reform Commission, 2023; Shanghai Statistics Bureau, 2026). The average published in-state tuition for public four-year colleges in the United States was $11,950, or about 17.8% of the U.S. per capita disposable personal income. Private nonprofit four-year tuition averaged $45,000 and was about 66.9% of the same U.S. income measure (College Board, 2025; U.S.) Bureau of Economic

Analysis, 2026. They are simple calculations in different data environments, and the impact on the finances of Chinese and American families is not shown directly.

Figure 3

Exploratory Tuition-to-Income Ratios in Separate Data Settings



Note. Calculated from Shanghai tuition policy, Shanghai resident income, College Board tuition, and U.S. BEA income data. Measures are not directly comparable across countries.

University Fee Examples in Shanghai

The earlier figures for China came from government policies. To see what the students can do, I also looked at the announcements from two universities. Fudan University's 2025 undergraduate admissions information shows a base tuition of 6,500-8,140 yuan per year and housing costs of about 1,200 yuan per year for a dorm room (Fudan University, 2025). Shanghai University's 2025 information lists tuition fees of 6,500 yuan for several humanities and social sciences programmes and 7,000 yuan for many science and engineering programmes. The yearly dorm fee will not exceed 1,200 yuan. Some special or joint programmes are more expensive (Shanghai University, 2025a, 2025b).

Table 3

Published Undergraduate Fees at Two Shanghai Universities, 2025

University and source year	Published annual tuition	Published dorm charge	Important note
Fudan University, 2025	CNY 6,500–8,140	About CNY 1,200	Tuition varies by program; room charge depends on actual housing
Shanghai University, 2025	CNY 6,500 (selected fields); CNY 7,000 (many science and engineering fields)	Up to CNY 1,200	Program examples only; special and joint programs can cost more

Note. Information is from Fudan University (2025) and Shanghai University (2025a, 2025b). Fees vary by program and housing arrangement.

These are the particular cases covered by the policy; not all of a student's living expenses are covered. Food, books, transportation and aid are not in the US figures and are not measured in the same way. Fudan's Housing Area is small; Shanghai University's is relatively large. For this reason, I will not use these examples to show that one country's colleges are cheaper.

Discussion

What the Results Support

First, the U.S. published tuition data for many schools. Average in-state tuition at public four-year colleges is about 2.9 times the public two-year amount, and the private nonprofit four-year average is about 10.8 times the public two-year amount (College Board, 2025). These figures are from the same report and school year; therefore, it will be more reasonable to compare them with the numbers from two different countries. They are still averages and do not cover all colleges.

Second, the selected Cost Measure will change the circumstances. Public two-year colleges are relatively low-cost in terms of tuition alone. But with the addition of estimated housing and food, the amount is now $15,000. Their tuition advantage has not disappeared, but the scope of funds is smaller. Students living at home have lower housing expenses; therefore, future research should consider various living arrangements.

A Simple Sensitivity Check

A small modification to the components in a calculation can be made to observe the resulting changes. Tuition for public two-year colleges is only $4,150, and adding the estimated costs of housing and food brings the total to about $15,000 (3.61 times higher). The amount for in-state students at

public four-year colleges increased from $11,950 to $25,850, an increase of about 2.16 times. It will not be paid by others. It shows that a claim about 'college expenses' needs to list the specific items of expenditure.

A second check concerns residency status. In the same College Board report and school year, the average public four-year tuition for out-of-state students was $31,880, about 2.67 times that of in-state students ($11,950). Given the substantial impact of student eligibility in a single country, it should not be disregarded in comparisons among countries.

Why I Cannot Rank China and the United States Yet

Only the Fudan and Shanghai University examples have been introduced for the tuition cap in some areas of Shanghai. Figures in the United States are national averages for each college type. The Sources of Income vary. We do not have directly comparable estimates of housing, food, financial aid and the distribution of household income for students in both countries. Therefore, Figure 3 cannot show which country has more affordable college education.

I wanted to know whether the level of national development, population and income could be considered. The problems need to be investigated; however, the current dataset is too small and uneven for testing. The two countries did not have to be related to the problem. More sites and more universities are needed for the extended experiment to build a stable statistical model.

Limitations and Next Steps

The other is lack of data coverage. Shanghai's tuition cap and the two university cases are not representative of all of China. The U.S. college-age average is not representative. Published tuition is not the amount after grant subsidies. The list of dormitory fees from Shanghai University does not include all living expenses in the US that have been estimated. The actual expenses of living on campus,

off-campus rental, or living at home will also differ. Finally, per capita income is not the total money in a single household, and the sources are not all from the same year.

In future research, I would select similar public universities in several Chinese and U.S. cities and collect tuition, housing, food, transportation, and available aid for the same academic year. Then I will create sample budgets for local and non-local students, as well as for families of different incomes. Therefore, the comparison will be closer to the actual situation of the students and will not assume that all families have the same resources.

Conclusion

I started this project because I went to school in China before coming to the United States for college. I wish to know the reasons for the different college expenses in these two countries. It turned out that the cost of living in America couldn't be directly converted from Chinese yuan. Institution type, place of residence and cost of living also affect published fees in the US. Housing and food account for more than 70% of the combined tuition, housing and food expenses for public two-year colleges in the figures above. These results are consistent with the data.

However, the current evidence does not support an equitable national ranking of college affordability in China and the United States. For a fair comparison, we need to know more about tuition fees, the actual cost of living for students after financial aid in both countries, and their families' incomes. I have learned that the cost of living for college students is not only tuition fees. I have learned that I need to know the above figures for the comparison.

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